Supplementary Information for: High Helium Retentivity in Goethite under Hydrothermal Conditions Measured by Step-Heat Experiments in Silica Glass Ampoules

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after editing informed method 3He reduction in supplement, (removing 4He clutter), then port it all here.

1. Analytical Methods for Hydrothermal Goethite

1.1. Uncertainty Quantification for Bounding Method

Helium-in-Goethite Diffusivity Inversions. Apparent values of D/a^2 for 3He and 4He were calculated from times and cumulative measured yields using the equations of Fechtig and Kalbitzer (1966). Lower and upper bounds for diffusivity were calculated using the upper bound for the helium contents of the ampoules before and after each step, evaluated using (**??**). Diffusivity for step i is a function of the cumulative fractional yield for step i $\Sigma^i F$ and the cumulative fractional yield of the previous step $\Sigma^{i-1} F$. The diffusivity is maximized when the difference between the fractional yields is largest, and minimized when the difference is smallest. This difference is maximized for the extreme case where the previous step ended with no permeation lag, and the current step ended with the maximum amount of permeation lag ($^i_{SS}n_t$), whereas the opposite is true for the minimum diffusivity.

$$
\begin{aligned}
{}^{UB}D_\omega &= D\Big(\Sigma^{i-1}F_m,\ \Sigma^i F_m + {}^i_{SS}n_t\Big) \\
{}^{LB}D_\omega &= D\Big(\Sigma^{i-1}F_m + {}^{i-1}_{SS}n_t,\ \Sigma^i F_m\Big)
\end{aligned}
\tag{1}
$$

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For the 3He diffusivities, a linear regression was fit to the steps with cumulative 3He fractional yield ≥7% (12 steps, Fig. **??**), yielding an R^2 statistic of 0.989. The fit's slope and intercept are used as a reference line to calculate "Delta" values for the diffusivities of 3He and 4He (Fig. **??**). No uncertainty on the total amounts of 3He and 4He is propagated into the diffusivities because the experiment operated at low cumulative yields where this error affects all diffusivities approximately homogeneously, so it is only relevant for the y-intercept of the regression and the corresponding calculated closure temperature. The uncertainty from the intercept and slope parameters of the linear regression is significantly larger than the uncertainty related to the total yield, so it is neglected.

Helium Isotope Ratio Uncertainty. The upper and lower bounds of the $\frac{^4He}{^3He}$ isotope ratios for each heating step are taken as the maximum and minimum $\frac{^4He}{^3He}$ isotope ratio calculated out of the 4 hypothetical end-member permeation scenarios, which would have been experienced by both helium isotopes systems:

1. $n_t = 0,\ n_0 = 0$
2. $n_t = n_t^{\mathrm{UB}},\ n_0 = 0$
3. $n_t = n_t^{\mathrm{UB}},\ n_0 = n_0^{\mathrm{UB}}$
4. $n_t = 0,\ n_0 = n_0^{\mathrm{UB}}$

Intuitively, higher differential permeability between the two helium isotopes ($\delta = \frac{K_{3He}}{K_{4He}}$ here) will result in greater isotope fractionation between the measured $\left(\frac{m_4}{m_3}\right)$ and true $\left(\frac{\omega_4}{\omega_3}\right)$. Furthermore, lower absolute permeabilities will result in stronger permeation lag, additionally producing higher isotope fractionation in the measured value. The 4He permeability is thus conservatively taken to be identical to Swets et al. (1961), an anhydrous literature value which this experiment's own isothermal 3He permeation measurements show to underestimate the ampoule's true permeability (Fig. **??**); 4He permeability itself could not be independently measured in this experiment (supplement, **??**). For the purposes of evaluating the measured isotope ratios, the relative 3He/4He permeability (δ) is taken to be 1.29, the 2σ upper bound of the value measured independently for a water-filled ampoule in Chapter **??** (Thyer et al., 2026): 1.10 ± 0.19 (2σ). 1.29 is in great excess of realistic values of the relative 3He/4He permeability, which is expected to be between 1.06-1.11 for the temperature range relevant to this experiment (Jones, 1953).

These conservative assumptions taken together guarantee that the calculated upper and lower bounds for the isotope ratios accurately constrain the true value. The upper bounds of the ampoule contents are calculated using the steady-state approximation of Eq. (**??**) (Chapter **??**): $n_{ss} = \frac{m}{\lambda t - 1}$. For the four extreme cases listed above, the following equations apply:

1. $\frac{\omega_4}{\omega_3} = \frac{m_4}{m_3}$
2. $\frac{\omega_4}{\omega_3} = \frac{m_4 + \frac{m_4}{\lambda t - 1}}{m_3 + \frac{m_3}{\delta\lambda t - 1}}$
3. $\frac{\omega_4}{\omega_3} = \frac{m_4 + \frac{m_4}{\lambda t - 1} - \frac{m_4^{i-1}}{(\lambda t)^{i-1} - 1}}{m_3 + \frac{m_3}{\delta\lambda t - 1} - \frac{m_3^{i-1}}{\delta(\lambda t)^{i-1} - 1}}$
4. $\frac{\omega_4}{\omega_3} = \frac{m_4 - \frac{m_4^{i-1}}{(\lambda t)^{i-1} - 1}}{m_3 - \frac{m_3^{i-1}}{\delta(\lambda t)^{i-1} - 1}}$

Where m_x^{i-1} and $(\lambda t)^{i-1}$ refer to the measured helium content and λt of the previous step, respectively. λ here applies explicitly to 4He, whereas $\delta\lambda$ applies to 3He. The upper and lower bounds of the isotope ratios are then taken to be the maximum and minimum of these four extreme cases.

1.2. Finite Difference Diffusion Model of Cooling Tube

One potential concern about the results of the internally measured permeabilities is that some additional helium was permeated during the cooling period after each permeation step. In order to evaluate the significance of this effect, a finite difference model was built in python to simulate diffusion, influx, and outflux of helium across a hollow glass cylinder containing an arbitrary amount of helium gas within a helium-free chamber using the procedure described in (Thyer et al., 2026). The glass cylinder was given the same dimensions of the ampoule used in the Win-06-01b hydrothermal experiment, and was subjected to the same heating and cooling schedule as the hydrothermal experiment's permeation steps. The tube was charged with 100 arbitrary units of helium atoms at the end of each experimental step and a steady state concentration profile was assumed for the new arbitrary charge at T = 24°. Permeability values for 3He were assumed as the product

of the 4He permeability values given by Swets et al. (1961) and the 3He/4He relative permeabilities given by Jones (1953).

To simulate cooling, a cooling rate had to be assumed. To most accurately simulate the cooling of the experimental ampoule, Watlow temperature records of the experimental ampoule during cooling steps were used to calculate the cooling rate dT/dt for discrete time intervals. A quadratic function was fit to these data points to model the cooling rate as a function of temperature with a single root at T = 24°C such that the temperature would stabilize at T = 24°C by design.

$$\frac{dT}{dt} = -\left(9.23081 \times 10^{-6}\right)(T-24)^2 - \left(7.74304 \times 10^{-4}\right)(T-24)$$

(dT/dt in °C/s, T in °C; $R^2 = 0.9633$).

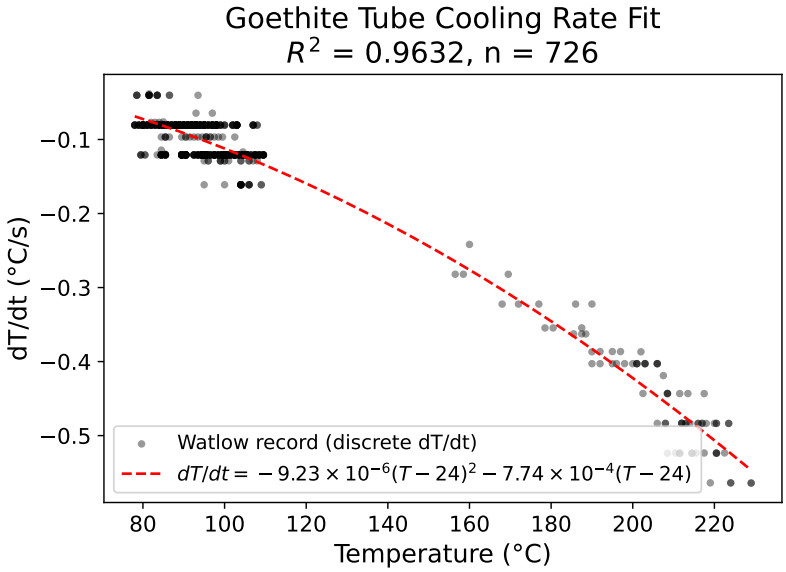


Figure S1: Discrete cooling rate (dT/dt) values calculated from the Watlow temperature record for the Winsor Goethite experiment, plotted against temperature, with the quadratic fit used to simulate cooling in the finite difference diffusion model overlain.

2. Encapsulated Durango apatite step-heat data

The following table gives the complete step-heat dataset for the silica glass encapsulated Durango apatite experiment described in §?? and plotted

in Fig. **??**. Permeation lag corrections follow the permeation-step route of §**??**: Eq. (**??**) is applied to the permeation steps shown in **bold** to solve for the ampoule's helium content, and Eq. (**??**) propagates that content forward and backward in time to the neighbouring experimental steps.

Only the last of each sequence of room-temperature measurements was used to solve for n_0, because the temperature records show that the ampoule had not yet cooled to room temperature when the first permeation step of each sequence began. Each pair of permeation steps nonetheless yielded values of n_0 within 2σ of one another.

Measurement uncertainties are estimated from the standard deviations of sensitivities and blanks determined by standards, and are propagated through Eqs. (**??**) and (**??**) in quadrature, together with a general 10% uncertainty on the use of Eq. (**??**). Uncertainty on a step's cumulative fractional yield is the sum of the uncertainty on its own fractional yield and those of all preceding steps, added in quadrature. The total yield is estimated from the measured mass of encapsulated apatite (3.18 mg) and its helium concentration of 194.2 nccSTP/mg (Farley, 2000); no uncertainty on the total yield is propagated into the reported diffusivities.

Table S1: Step heat data for the silica glass encapsulated Durango apatite experiment. $^{He}F_{Tube}$ is the fractional permeation yield (Eq. (**??**)) of the bracketing permeation step; $^{He}F_{Apatite}$ is the permeation lag-corrected cumulative fractional yield of helium released from the apatite. n_0 values (initial ^{4}He content of the ampoule, Eq. (**??**)) are shown in **bold** for permeation steps used directly to solve for n_0, and in *italics* for all other steps, where n_0 is instead propagated forward or backward in time from the nearest bolded permeation step using Eq. (**??**). Blank "Type" entries denote room-temperature measurements preceding the permeation step used to solve for n_0, taken before the diffusive steady state concentration profile was reestablished (see text). Uncertainties are 2σ. Corrected $\ln(D/a^2)$ values shown as "–" correspond to steps where the corrected apatite yield (ω) was within 2σ of 0 and were excluded from the Arrhenius plot (Fig. **??**).

Type	Time (min)	Temp. (°C)	m (nccSTP ^{4}He)	$^{He}F_{Tube}$	n_0 (nccSTP ^{4}He)	ω (nccSTP ^{4}He)	$^{He}F_{Apatite}$	Corrected $\ln(D/a^2)$	Uncorrected $\ln(D/a^2)$
Permeation	15	24.3	0.00704	± 4.76‰	**1.48**	±			
			0.00046		**0.18**				
Experimental	240	100.0	0.1700 ± 0.0011		*1.47*	± 0.05	± 0.00803%	–	-28.39
					0.18	0.24			
	15	28.0	0.00721	±	*1.35*	±			
			0.00039		*0.16*				
	15	24.0	0.00639	±	*1.34*	±			
			0.00039		*0.16*				
Permeation	15	24.0	0.00630	± 4.71‰	**1.34**	±			
			0.00039		**0.16**				
Experimental	240	150.0	0.3090 ± 0.0023		*1.33*	± 0.01	± 0.00983%	–	-26.98
					0.16	0.21			
	15	28.1	0.00542	±	*1.03*	±			
			0.00039		*0.13*				
	15	24.3	0.00484	±	*1.03*	±			
			0.00039		*0.13*				

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Type	Time (min)	Temp. (°C)	m (nccSTP ^{4}He)		$^{He}F_{Tube}$	n_0 (nccSTP ^{4}He)		ω (nccSTP ^{4}He)	$^{He}F_{Apatite}$	Corrected $\ln(D/a^2)$	Uncorrected $\ln(D/a^2)$
Permeation	15	24.2	0.00485	±	4.73‰	**1.02**	±				
			0.00039			**0.13**					
Experimental	240	200.1	0.9416 ± 0.0085			*1.02*	±	1.64	± 0.276%	-23.80	± -25.02
						0.13		0.23		0.28	
	15	29.9	0.00947	±		*1.72*	±				
			0.00041			*0.19*					
	15	23.8	0.00786	±		*1.71*	±				
			0.00040			*0.19*					
Permeation	15	24.0	0.00804	±	4.72‰	**1.70**	±				
			0.00040			**0.19**					
Experimental	60	250.2	1.097 ± 0.011			*1.69*	±	3.70	± 0.874%	-20.21	± -24.01
						0.19		0.48		0.20	
	15	33.0	0.02653	±		*4.30*	±				
			0.00050			*0.44*					
	15	23.5	0.02016	±		*4.27*	±				
			0.00046			*0.44*					
Permeation	15	24.0	0.02003	±	4.72‰	**4.25**	±				
			0.00047			**0.44**					
Experimental	60	270.3	2.508 ± 0.027			*4.23*	±	5.26	± 1.73%	-19.04	± -22.51
						0.44		0.83		0.22	
	15	33.9	0.04414	±		*6.98*	±				
			0.00067			*0.70*					

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Type	Time (min)	Temp. (°C)	m (nccSTP ^{4}He)	$^{He}F_{Tube}$	n_0 (nccSTP ^{4}He)	ω (nccSTP ^{4}He)	$^{He}F_{Apatite}$	Corrected $\ln(D/a^2)$	Uncorrected $\ln(D/a^2)$
Permeation	15	23.6	0.03246 ± 0.00057	± 4.68‰	**6.93 ± 0.70**				
Experimental	10	290.3	4.979 ± 0.058		*6.90 ± 0.70*	9.0 ± 1.3	3.18%	-17.87 ± 0.20	-21.10
	15	33.6	0.06816 ± 0.00095		*10.9 ± 1.1*				
Permeation	15	23.6	0.05050 ± 0.00078	± 4.66‰	**10.8 ± 1.1**				
Experimental	10	300.3	7.243 ± 0.091		*10.8 ± 1.1*	9.5 ± 1.7	4.72%	-17.34 ± 0.25	-20.07
	15	34.2	0.0843 ± 0.0012		*13.1 ± 1.3*				
	15	23.5	0.06160 ± 0.00094		*13.0 ± 1.3*				
Permeation	15	24.1	0.06097 ± 0.00095	± 4.72‰	**12.9 ± 1.3**				
Experimental	10	285.3	5.919 ± 0.080		*12.9 ± 1.3*	3.6 ± 1.7	5.30%	-18.04 ± 0.71	-20.65
	15	36.8	0.0729 ± 0.0011		*10.5 ± 1.1*				
Permeation	15	23.6	0.04846 ± 0.00084	± 4.64‰	**10.4 ± 1.1**				
Experimental	10	275.2	5.382 ± 0.077		*10.4 ± 1.1*	2.5 ± 1.3	5.71%	-18.6 ± 1.1	-19.80
	15	37.0	0.05243 ± 0.00092		*7.53 ± 0.76*				

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Type	Time (min)	Temp. (°C)	m (nccSTP ^{4}He)		${}^{He}F_{Tube}$	n_0 (nccSTP ^{4}He)		ω (nccSTP ^{4}He)		${}^{He}F_{Apatite}$	Corrected $\ln(D/a^2)$	Uncorrected $\ln(D/a^2)$
Permeation	15	23.6	0.03487	±	4.66‰	**7.48**	±					
			0.00070			**0.76**						
Experimental	100	255.2	3.382 ± 0.052			*7.44*	±	0.60	±	5.81%	–	-20.22
						0.76		0.90				
	15	37.1	0.03206	±		*4.66*	±					
			0.00069			*0.48*						
Permeation	15	23.6	0.02147	±	4.64‰	**4.63**	±					
			0.00056			**0.48**						
Experimental	60	300.3	3.684 ± 0.059			*4.61*	±	6.68	±	6.89%	-17.17	± -20.76
						0.48		0.91			0.20	
	15	36.3	0.0518 ± 0.0010			*7.60*	±					
						0.77						
Permeation	15	23.6	0.03513	±	4.65‰	**7.55**	±					
			0.00077			**0.77**						
Experimental	120	300.2	9.78 ± 0.17			*7.51*	±	11.1	±	1.2 8.69%	-17.13	± -19.21
						0.77					0.22	
	15	35.7	0.0595 ± 0.0012			*8.85*	±					
						0.90						
Permeation	15	23.6	0.04087	±	4.65‰	**8.79**	±					
			0.00089			**0.90**						
Experimental	180	300.1	14.58 ± 0.27			*8.75*	±	13.8	±	1.2 10.9%	-17.08	± -18.75
						0.90					0.18	

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Type	Time (min)	Temp. (°C)	m (nccSTP ^{4}He)	$^{He}F_{Tube}$	n_0 (nccSTP ^{4}He)	ω (nccSTP ^{4}He)	$^{He}F_{Apatite}$	Corrected $\ln(D/a^2)$	Uncorrected $\ln(D/a^2)$
	15	35.9	0.0547 ± 0.0012		*7.96 ± 0.81*				
Permeation	15	23.6	0.03782 ± 0.00089	± 4.78‰	**7.91 ± 0.81**				
Experimental	10	320.3	7.49 ± 0.14		*7.87 ± 0.81*	11.4 ± 1.5	12.8%	-15.96 ± 0.18	-17.94
	15	36.9	0.0813 ± 0.0017		*11.7 ± 1.2*				
Permeation	15	23.5	0.0541 ± 0.0012	4.64‰	**11.7 ± 1.2**				
Experimental	10	330.4	11.18 ± 0.22		*11.6 ± 1.2*	14.5 ± 1.9	15.1%	-15.54 ± 0.21	-18.08
	15	37.4	0.1076 ± 0.0023		*14.9 ± 1.5*				
Permeation	15	23.7	0.0691 ± 0.0016	4.67‰	**14.8 ± 1.5**				
Experimental	10	350.4	19.78 ± 0.41		*14.7 ± 1.5*	28.3 ± 2.8	19.7%	-14.61 ± 0.15	-18.11
	15	37.0	0.1615 ± 0.0035		*23.3 ± 2.4*				
Permeation	15	23.5	0.1071 ± 0.0024	4.64‰	**23.1 ± 2.4**				
Experimental	10	370.4	32.97 ± 0.71		*23.0 ± 2.4*	41.3 ± 4.0	26.4%	-13.90 ± 0.15	-16.15
	15	37.3	0.2207 ± 0.0049		*31.4 ± 3.2*				

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Type	Time (min)	Temp. (°C)	m (nccSTP ^{4}He)	$^{He}F_{Tube}$	n_0 (nccSTP ^{4}He)	ω (nccSTP ^{4}He)	$^{He}F_{Apatite}$	Corrected $\ln(D/a^2)$	Uncorrected $\ln(D/a^2)$
Permeation	15	23.5	0.1447 ± 0.0033	4.65‰	**31.1 ± 3.2**				
Experimental	10	385.4	42.32 ± 0.94		*31.0 ± 3.2*	44.8 ± 4.8	33.6%	-13.48 ± 0.17	-15.41
	15	37.8	0.2393 ± 0.0055		*33.5 ± 3.4*				
Permeation	15	23.5	0.1548 ± 0.0037	4.66‰	**33.2 ± 3.4**				
Experimental	10	360.4	27.53 ± 0.64		*33.1 ± 3.4*	15.3 ± 4.1	36.1%	-14.35 ± 0.51	-15.33
	15	39.6	0.1620 ± 0.0039		*20.9 ± 2.1*				
Permeation	15	23.6	0.0963 ± 0.0024	4.65‰	**20.7 ± 2.1**				
Experimental	10	340.4	14.78 ± 0.35		*20.6 ± 2.1*	7.4 ± 2.6	37.3%	-15.00 ± 0.90	-15.63
	15	40.0	0.1029 ± 0.0026		*13.2 ± 1.4*				
Permeation	15	23.5	0.0610 ± 0.0017	4.64‰	**13.1 ± 1.4**				
Experimental	100	325.2	12.37 ± 0.31		*13.1 ± 1.4*	6.8 ± 1.6	38.4%	-15.6 ± 1.1	-13.69
	15	40.1	0.0589 ± 0.0017		*7.50 ± 0.78*				
Permeation	15	24.0	0.0351 ± 0.0011	4.72‰	**7.44 ± 0.78**				

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Type	Time (min)	Temp. (°C)	m (nccSTP ^{4}He)	$^{He}F_{Tube}$	n_0 (nccSTP ^{4}He)	ω (nccSTP ^{4}He)	$^{He}F_{Apatite}$	Corrected $\ln(D/a^2)$	Uncorrected $\ln(D/a^2)$
Experimental	240	400.1	101.1 ± 2.7		*7.40 ± 0.78*	110.0 ± 3.2	56.2%	-13.281 ± 0.076	-13.70
	15	34.9	0.1096 ± 0.0031		*16.3 ± 1.7*				
Permeation	15	23.8	0.0757 ± 0.0022	4.67‰	**16.2 ± 1.7**				
Experimental	10	350.3	14.88 ± 0.41		*16.1 ± 1.7*	5.8 ± 1.9	57.2%	-14.92 ± 0.92	-15.42
	15	39.5	0.0567 ± 0.0017		*7.06 ± 0.74*				
Permeation	15	24.0	0.0330 ± 0.0011	4.71‰	**7.00 ± 0.74**				
Experimental	60	400.4	14.21 ± 0.40		*6.97 ± 0.74*	18.1 ± 1.4	60.1%	-13.31 ± 0.31	-15.50
	15	40.4	0.0866 ± 0.0026		*10.9 ± 1.1*				
Permeation	15	24.0	0.0508 ± 0.0016	4.71‰	**10.8 ± 1.1**				
Experimental	120	300.2	7.65 ± 0.22		*10.7 ± 1.1*	0.7 ± 1.2	60.2%	–	-16.50
	15	39.4	0.0291 ± 0.0011		*3.79 ± 0.41*				
Permeation	15	23.7	0.01758 ± 0.00077	4.68‰	**3.76 ± 0.41**				

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